Capstone Project Report
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TRIPLE POWER LAW FIT ANALYSIS RESULTS
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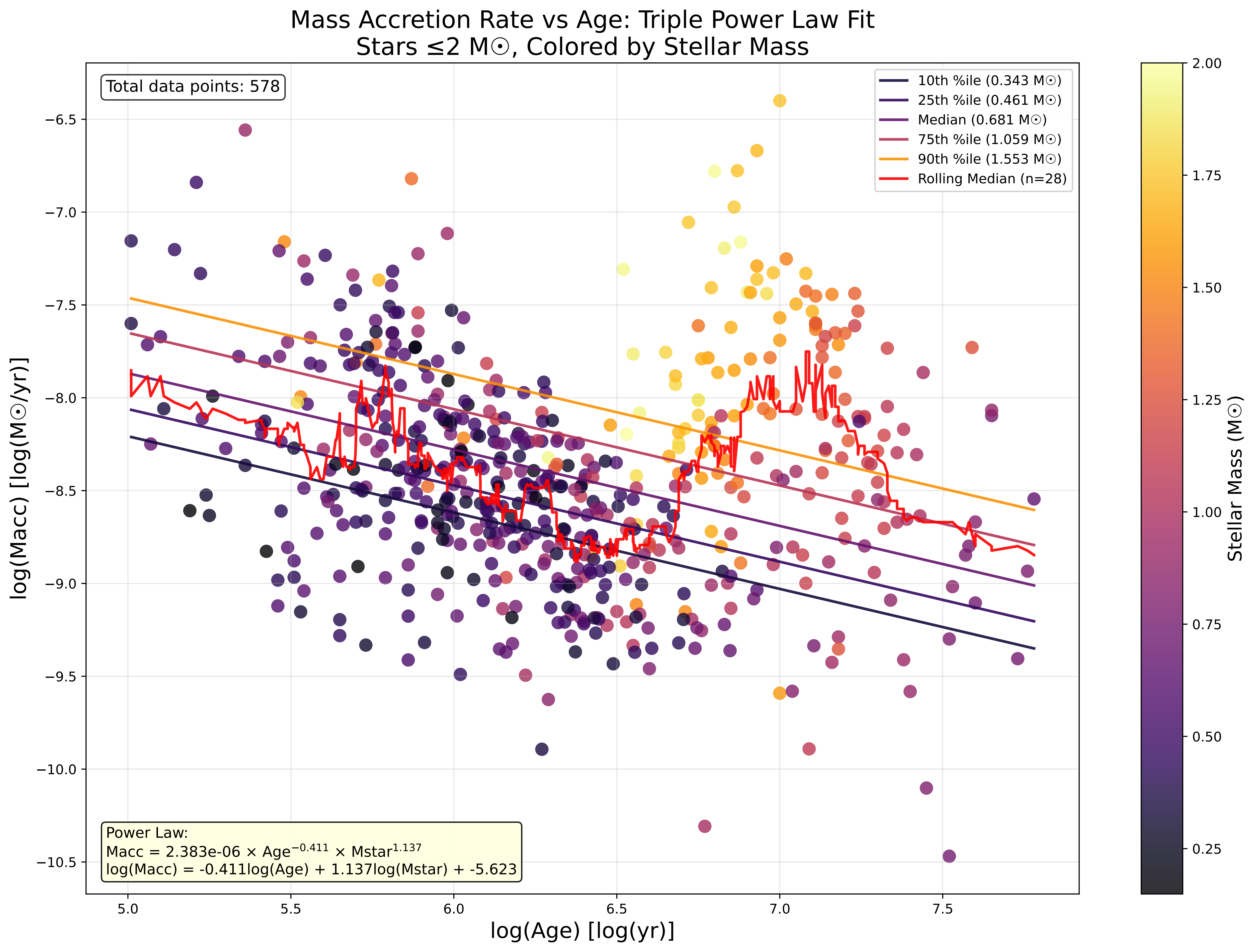
DATA STATISTICS:
Total data points: 578
logMacc range: -10.469 to -6.400
logAge range: 5.010 to 7.780
Mstar range: 0.149 to 1.991 Solar Masses

CORRELATIONS:
logAge vs logMacc: -0.139
logMstar vs logMacc: 0.249
logAge vs logMstar: 0.547

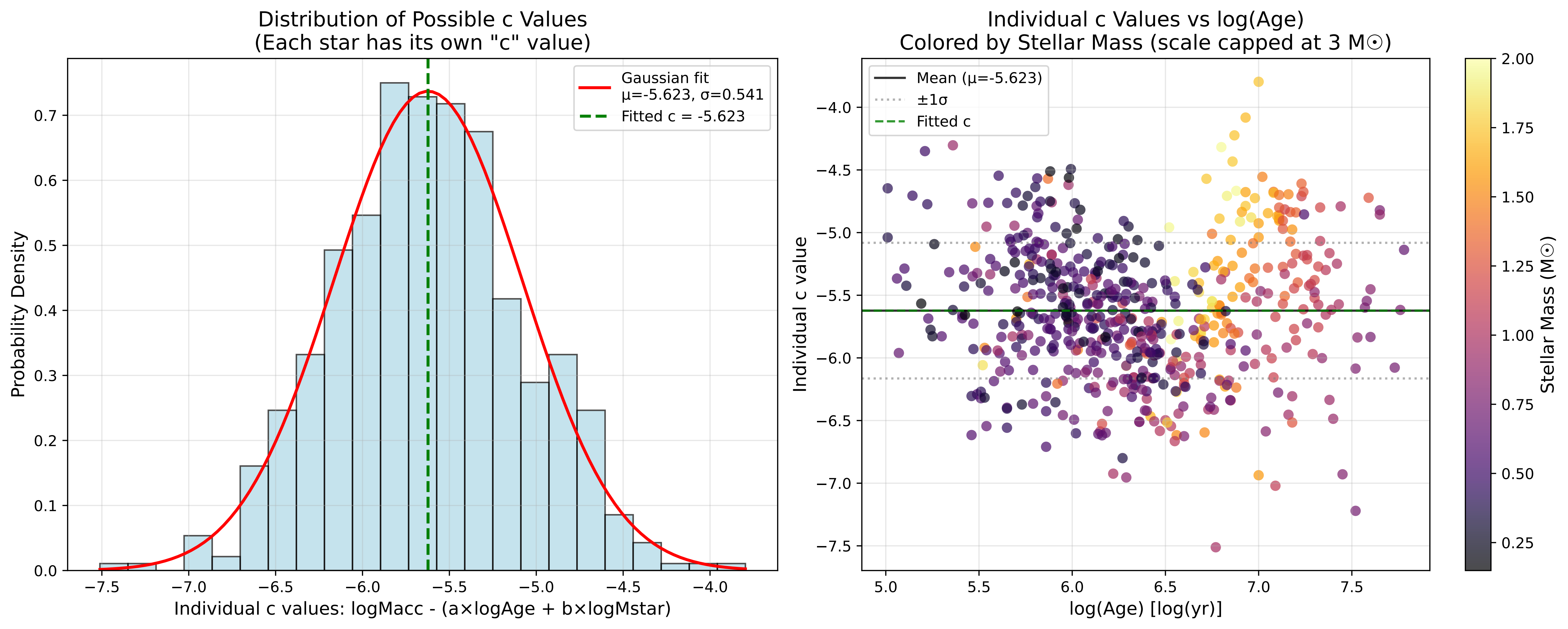
TRIPLE POWER LAW FIT RESULTS:
$\log(\text{Macc}) = -5.623 + -0.411 \times \text{logAge} + 1.137 \times \log(\text{Mstar})$
$\text{Macc} = 2.383\text{e-}06 \times \text{Age}^{-0.411} \times \text{Mstar}^{1.137}$

INTERPRETATION:
→ Mass accretion decreases with age (expected)
→ More massive stars have higher accretion rates (expected)

PROBABILISTIC ANALYSIS:
Individual c values distribution: $c \sim N(\mu=-5.623, \sigma=0.541)$
→ Intrinsic scatter in accretion physics: $\sigma = 0.541$
→ 68% of stars have c values within ±0.541 of mean
→ Age & mass explain 17.038% of variance
→ Unexplained variance: 82.818%



Mass Accretion Rate vs Age: Triple Power Law Fit
Stars ≤2 M☉, Colored by Stellar Mass
Total data points: 578
10th %ile (0.343 M☉)
25th %ile (0.461 M☉)
Median (0.681 M☉)
75th %ile (1.059 M☉)
90th %ile (1.553 M☉)
Rolling Median (n=28)
Power Law:
Macc = 2.383e-06 × Age^−0.411 × Mstar^1.137
log(Macc) = -0.411log(Age) + 1.137log(Mstar) + -5.623
log(Macc) [log(M☉/yr)]
log(Age) [log(yr)]
Stellar Mass (M☉)
−6.5
−7.0
−7.5
−8.0
−8.5
−9.0
−9.5
−10.0
−10.5
5.0
5.5
6.0
6.5
7.0
7.5
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25



Distribution of Possible c Values
(Each star has its own "c" value)
Gaussian fit
μ=-5.623, σ=0.541
Fitted c = -5.623
Probability Density
Individual c values: logMacc - (a×logAge + b×logMstar)
Individual c Values vs log(Age)
Colored by Stellar Mass (scale capped at 3 M☉)
Mean (μ=-5.623)
±1σ
Fitted c
Individual c value
log(Age) [log(yr)]
Stellar Mass (M☉)